JavaScript Classes

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ECMAScript 2015, also known as ES6, introduced JavaScript Classes.

JavaScript Classes are templates for JavaScript Objects.

JavaScript Class Syntax

Use the keyword `class` to create a class.

Always add a method named `constructor()`:

Syntax

```
class ClassName {
  constructor() { ... }
}
```

Example

```
class Car {
  constructor(name, year) {
    this.name = name;
    this.year = year;
  }
}
```

A JavaScript class is **not** an object.

It is a **template** for JavaScript objects.

Using a Class

When you have a class, you can use the class to create objects:

Example

```
const myCar1 = new Car("Ford", 2014);
const myCar2 = new Car("Audi", 2019);
```

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The example above uses the **Car class** to create two **Car objects**.

The constructor method is called automatically when a new object is created.

The Constructor Method

The constructor method is a special method:

- It has to have the exact name "constructor"
- It is executed automatically when a new object is created
- It is used to initialize object properties

If you do not define a constructor method, JavaScript will add an empty constructor method.

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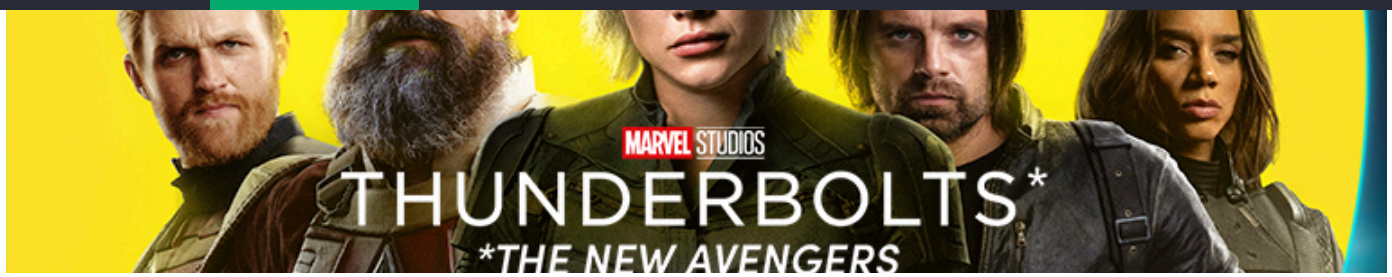

Class Methods

Class methods are created with the same syntax as object methods.

Use the keyword `class` to create a class.

Always add a `constructor()` method.

Then add any number of methods.

Syntax

```
class ClassName {
  constructor() { ... }
  method_1() { ... }
  method_2() { ... }
  method_3() { ... }
}
```

Create a Class method named "age", that returns the Car age:

Example

```
class Car {
  constructor(name, year) {
    this.name = name;
    this.year = year;
  }
  age() {
    const date = new Date();
    return date.getFullYear() - this.year;
  }
}
```

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You can send parameters to Class methods:

Example

```
class Car {
  constructor(name, year) {
    this.name = name;
    this.year = year;
  }
  age(x) {
    return x - this.year;
  }
}

const date = new Date();
let year = date.getFullYear();

const myCar = new Car("Ford", 2014);
document.getElementById("demo").innerHTML=
"My car is " + myCar.age(year) + " years old.";
```

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"use strict"

Classes syntax must be written following The "use strict" Directive.

You will get an error if you do not follow the "strict mode" rules.

Example

In "strict mode" you will get an error if you use a variable without declaring it:

```
class Car {
  constructor(name, year) {
    this.name = name;
```

```
    const date = new Date(); // This will work
    return date.getFullYear() - this.year;
  }
}
```

Try it Yourself »

Exercise

There is one method all classes should have, which one?

- init()
- constructor()
- onerror()

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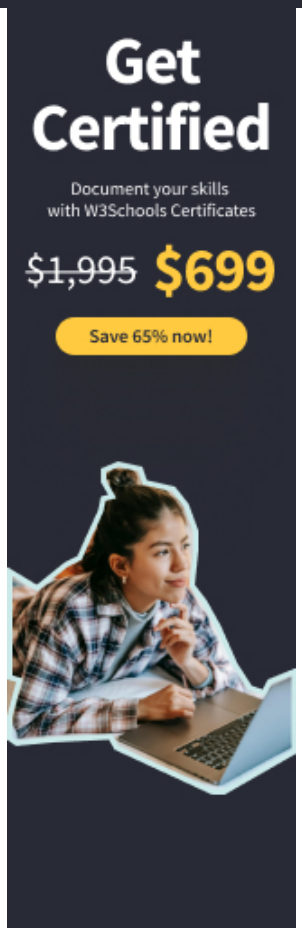

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